Score-Based Perspective of Diffusion Models

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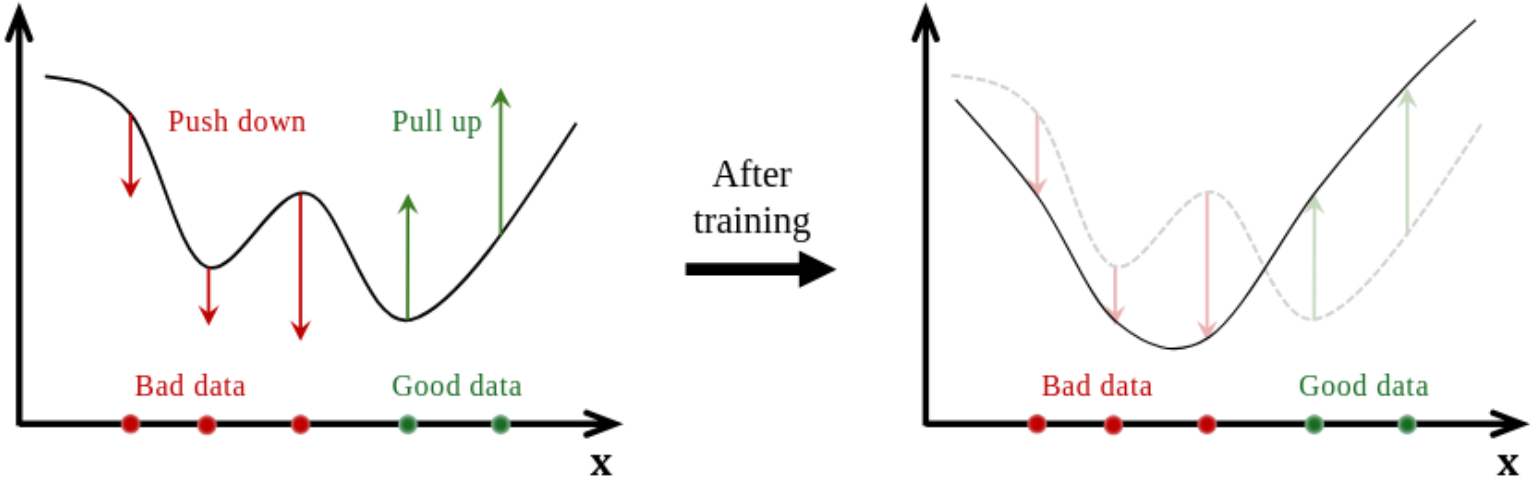
Energy-Based Models

Let $x \in \mathbb{R}^D$ denote a data point. EBMs define a probability density function via a parametric energy function $E_\phi(x)$:

$$p_\phi(x) := \frac{\exp(-E_\phi(x))}{Z_\phi}, \quad Z_\phi := \int_{\mathbb{R}_D} \exp(-E_\phi(x))dx. \tag{1}$$

Nice property: EBMs obey global trade-off.



Challenges of MLE in EBMs

EBMs can be trained by maximum likelihood, which balances fitting the data with global regularization:

$$\mathcal{L}_{MLE}(\phi) = \mathbb{E}_{p_{data}(x)}[\log \frac{\exp(-E_\phi(x))}{Z_\phi}] \tag{2}$$

$$= -\mathbb{E}_{p_{data}(x)}[E_\phi(x))] - \log Z_\phi \tag{3}$$

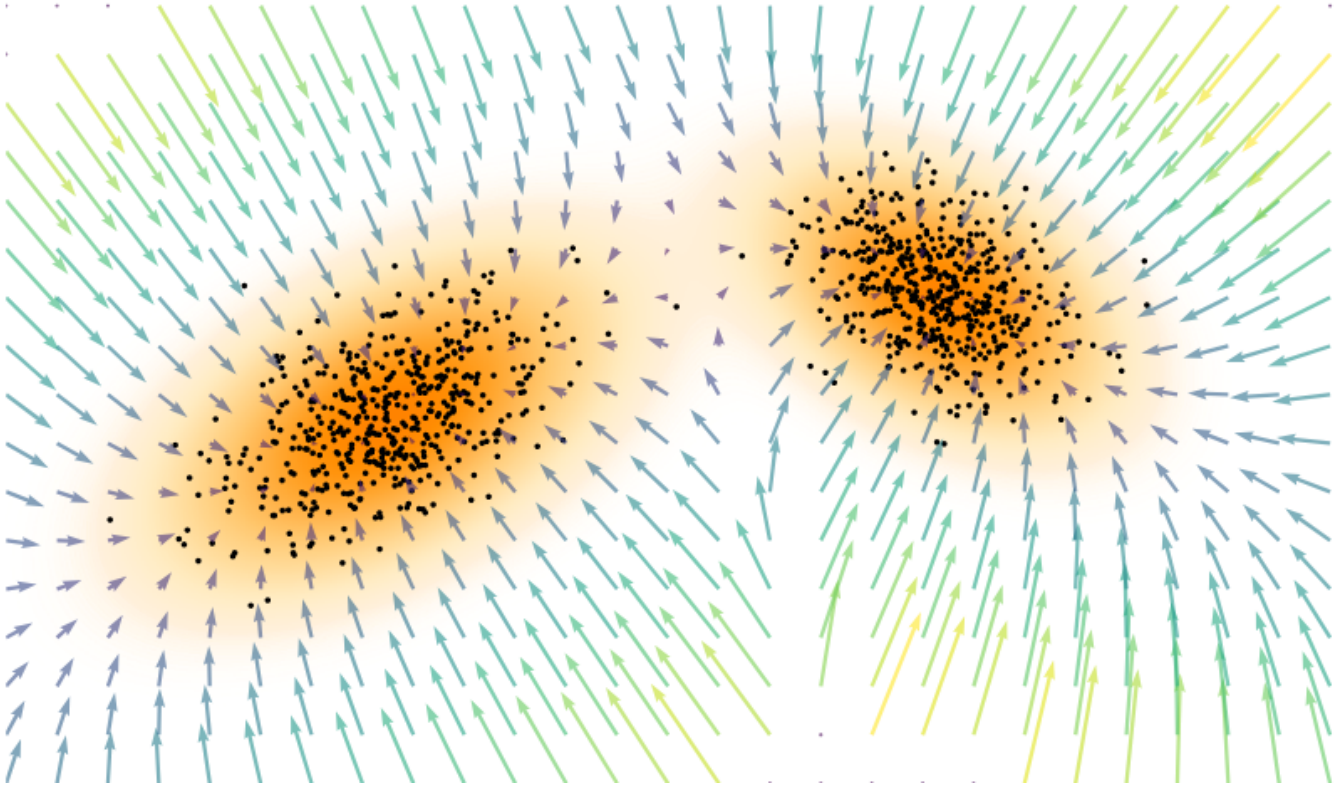
$$= -\mathbb{E}_{p_{data}(x)}[E_\phi(x))] - \log \int \exp(-E_\phi(x))dx. \tag{4}$$

However, in high dimensions computing $\log Z_\phi$ and its gradient is intractable, as it requires expectations under the model distribution.

Why Scores?

The score function is the gradient of the log-density:

$$s(x) := \nabla_x \log p(x). \tag{5}$$



Why Scores?

We can avoid learning the partition function. Many distributions are defined only up to an unnormalized density $\tilde{p}(x)$, e.g., $\exp(-E_\phi(x))$ in EBMs:

$$p(x) = \frac{\tilde{p}(x)}{Z}, \quad Z = \int \tilde{p}(x)dx. \tag{6}$$

The score only depends on $\tilde{p}$:

$$\nabla_x \log p(x) = \nabla_x \log \tilde{p}(x) - \nabla_x \log Z = \nabla_x \log \tilde{p}(x). \tag{7}$$

The distribution can be recoved from the score. Since it is the gradient of the log-density, the density can be recovered (up to a constant) via

$$\log p(x) = \log p(x_0) + \int_0^1 s(x_0 + t(x - x_0))^T (x - x_0) dt, \qquad (8)$$

where x_0 is a reference point and $\log p(x_0)$ is fixed by normalization. Thus, modeling the score is as expressive as modeling $p(x)$ itself.

How to train EBMs via Score Matching?

Score matching (Hyvärinen and Dayan, 2005) trains EBMs by aligning the model score with the (unknown) data score:

$$\mathcal{L}_{SM}(\phi) = \frac{1}{2}\mathbb{E}_{p_{data}(x)}||\nabla_x \log p_\phi(x) - \nabla_x \log p_{data}(x)||_2^2. \tag{9}$$

Although the data score is unknown, integration by parts yields an equivalent expression involving only the energy and its derivatives:

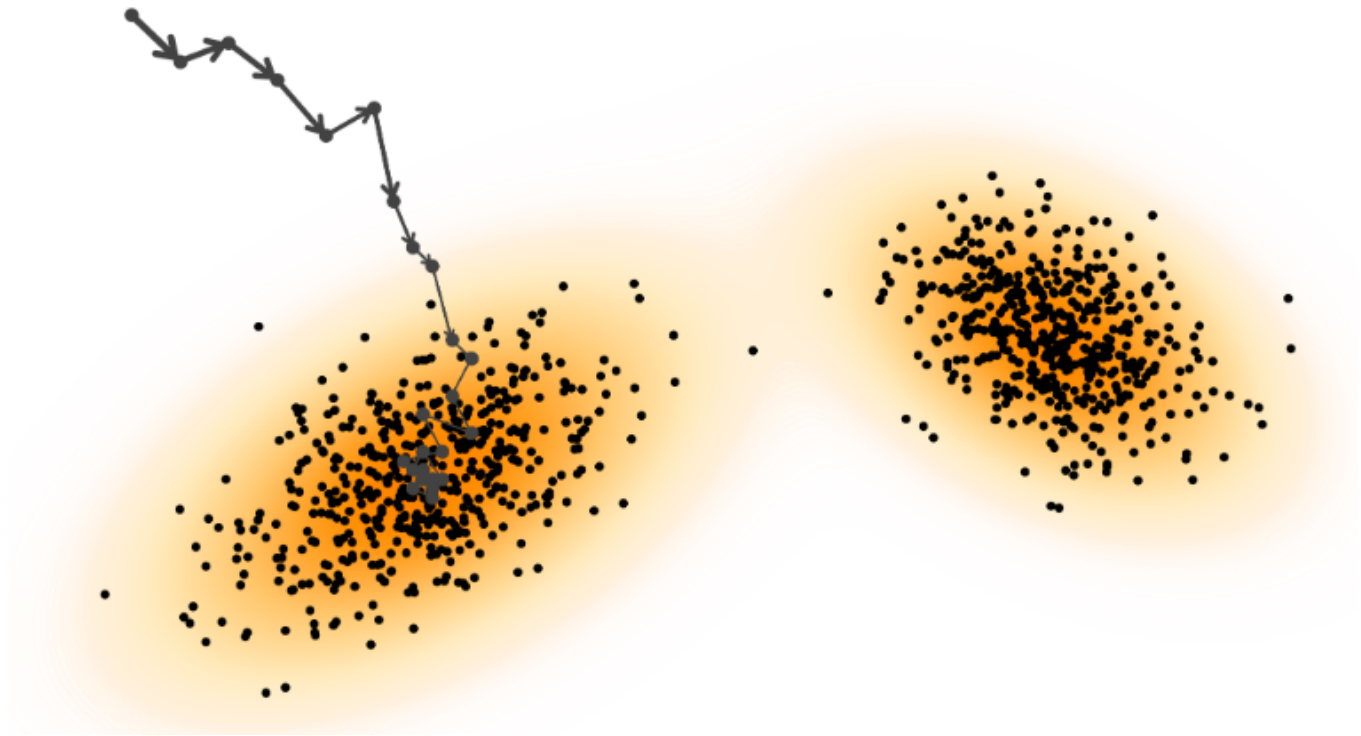
$$\mathcal{L}_{SM}(\phi) = \mathbb{E}_{p_{data}(x)}[-Tr(\nabla_x^2 \log p_\phi(x)) + \frac{1}{2}||\nabla_x \log p_\phi(x)||_2^2] + C. \tag{10}$$

How to sample EBMs?

Sampling from EBMs can be performed using Langevin dynamics. The discrete-time Langevin update is

$$x_{n+1} = x_n + \eta \nabla_x \log p_\phi(x_n) + \sqrt{2\eta}\epsilon_n, \tag{11}$$

where x_0 is initialized from some distribution, $\eta > 0$ is the step size, and $\epsilon_n \sim \mathcal{N}(0, I)$ is Gaussian noise.



Continuous-Time Langevin Dynamics

As the step size $\eta \to 0$, the discrete Langevin dynamics converge to a continuous-time process described by the Langevin Stochastic Differential Equation (SDE):

$$d\mathbf{x}(t) = \nabla_{\mathbf{x}} \log p_\phi(\mathbf{x}(t))dt + \sqrt{2}d\mathbf{w}(t), \tag{12}$$

where $\mathbf{w}(t)$ denotes standard Brownian motion. Under certain conditions, the distribution of $\mathbf{x}(t)$ converges to p_ϕ.

Issues with Langevin Dynamics

- Langevin dynamics faces serious limitations in high-dimensional spaces, where it often requires an extremely long time to transition between regions of high probability.
- Its efficiency is highly sensitive to the choice of step size η, noise scale, and the number of iterations required to approximate the target distribution accurately.

Score-Based Generative Models

Score matching fits this vector field by minimizing the mean squared error (MSE) between the true and estimated scores:

$$\mathcal{L}_{SM}(\phi) = \frac{1}{2}\mathbb{E}_{p_{data}(x)}||s_\phi(x) - s(x)||_2^2. \tag{13}$$

We can optimize this objective without knowing the true score by using an equivalent objective:

$$\tilde{\mathcal{L}}_{SM}(\phi) = \mathbb{E}_{p_{data}(x)}[Tr(\nabla_x s_\phi(x)) + \frac{1}{2}||s_\phi(x)||_2^2]. \tag{14}$$

The alternative score matching objective can be understood directly from its two terms:

- Since the expectation is taken under p_{data}, the norm term $||s_\phi(x)||_2^2$ suppresses the score in high-probability areas, making them stationary.
- The divergence term $Tr(\nabla_x s_\phi(x))$ favors negative values, so that stationary points act as attractive sinks.

For a stationary point x^*, where $s_\phi(x^*) = \nabla_x u(x^*) = 0$, a second order Taylor expansion gives

$$u(x) = u(x^*) + \frac{1}{2}(x - x^*)^T \nabla_x^2 u(x^*)(x - x^*) + o(||x - x^*||^2) \quad (15)$$

If the Hessian is negative definite, then u is locally concave at x^* and the log density attains a strict local maximum there. Because all eigenvalues of the Hessian are negative, the trace is also negative. Thus the learned vector field has negative divergence and the stationary point is a sink: small perturbations are contracted back toward x^*.

Quick Recap

First objective:

$$\mathcal{L}_{SM}(\phi) = \frac{1}{2}\mathbb{E}_{p_{data}(x)}||s_\phi(x) - s(x)||_2^2. \tag{16}$$

Issues: We do not know the value of $s(x)$.
Second objective:

$$\tilde{\mathcal{L}}_{SM}(\phi) = \mathbb{E}_{p_{data}(x)}[Tr(\nabla_x s_\phi(x)) + \frac{1}{2}||s_\phi(x)||_2^2]. \tag{17}$$

Issues: Although tractable, it requires computing the trace of the Jacobian, which has worst-case complexity $O(D^2)$.

Sliced Score Matching and Hutchinson's Estimator

(Song et al., 2020b) Let $u \in \mathbb{R}^D$ be an isotropic random vector (e.g., Rademacher or standard Gaussian) with $\mathbb{E}[u] = 0$ and $\mathbb{E}[uu^T] = I$. By Hutchinson's identity

$$Tr(A) = \mathbb{E}_u[u^T A u], \quad \mathbb{E}_u[(u^T b)^2] = ||b||_2^2, \tag{18}$$

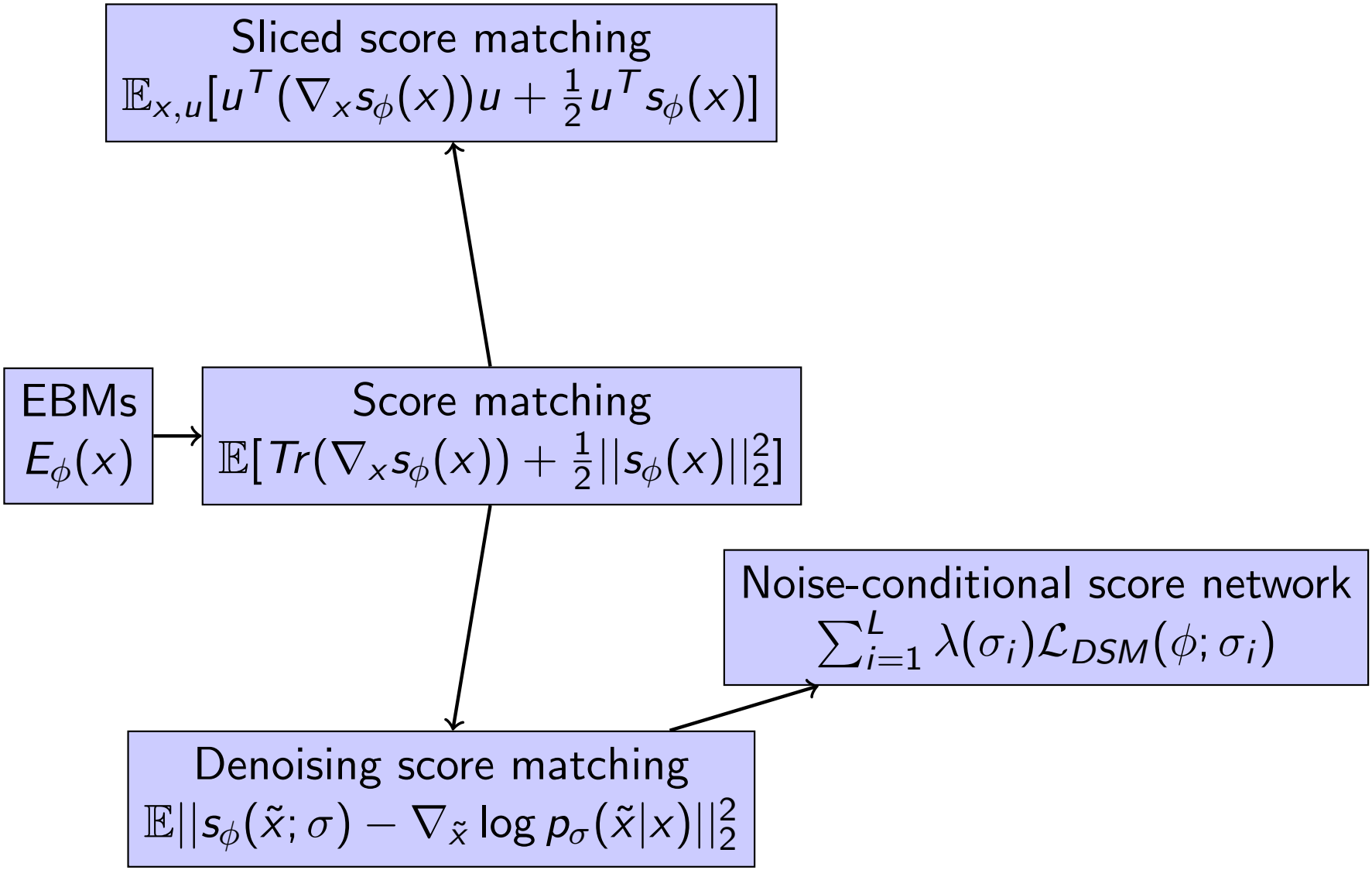
Hence, the score objective is equivalent to:

$$\tilde{\mathcal{L}}_{SM}(\phi) = \mathbb{E}_{p_{data}(x),u}[u^T(\nabla_x s_\phi(x))u + \frac{1}{2}u^T s_\phi(x)], \qquad (19)$$

For which the first term can be computed efficiently using JVP/VJP operations.

More issues: $\nabla_x \log p_{data}(x)$ may be undefined or unstable, and the method only constrains the vector field at observed points, providing weak control in their neighborhoods. It further suffers from probe-induced variance and repeated JVP/VJP costs.

Relationship diagram



Denoising Score Matching (DSM)

DSM addresses these issues by injecting noise into the data via a known distribution $p_\sigma(\tilde{x}|x)$. $s_\phi(\tilde{x};\sigma)$ is trained to approximate the score of the marginal perturbed distribution

$$p_\sigma(\tilde{x}) = \int p_\sigma(\tilde{x}|x) p_{data}(x) dx \tag{20}$$

by minimizing the loss

$$\mathcal{L}_{SM}(\phi;\sigma) = \frac{1}{2}\mathbb{E}_{p_\sigma}||s_\phi(\tilde{x};\sigma) - \nabla_{\tilde{x}} \log p_\sigma(\tilde{x})||_2^2 \tag{21}$$

Conditioning produces an equivalent, tractable objective:

$$\mathcal{L}_{DSM}(\phi;\sigma) = \frac{1}{2}\mathbb{E}_{p_{data},p_\sigma(\cdot|x)}||s_\phi(\tilde{x};\sigma) - \nabla_{\tilde{x}} \log p_\sigma(\tilde{x}|x)||_2^2 \quad (22)$$

For gaussian noise, we get the DSM loss:

$$\mathcal{L}_{DSM}(\phi;\sigma) = \frac{1}{2}\mathbb{E}_{x,\tilde{x}|x}||s_\phi(\tilde{x};\sigma) - \frac{x-\tilde{x}}{\sigma^2}||_2^2 \quad (23)$$

$$= \frac{1}{2}\mathbb{E}_{x,\epsilon}||s_\phi(x+\sigma\epsilon;\sigma) - \frac{\epsilon}{\sigma}||_2^2 \quad (24)$$

where $\epsilon \sim \mathcal{N}(0, I)$.

Advantages of Noise Injection:

- The noisy distribution has full support and its score function is well defined everywhere.
- The noise smooths out sparse regions between modes, allowing for Langevin dynamics to traverse low-density regions more effectively.

Relationship to Denoising

Tweedie's Formula Assume $x \sim p_{data}$ and $\tilde{x} \sim \mathcal{N}(\cdot; \alpha x, \sigma^2 I)$ then

$$\alpha \mathbb{E}_{x \sim p(\cdot|\tilde{x})}[x|\tilde{x}] = \tilde{x} + \sigma^2 \nabla_{\tilde{x}} \log p_\sigma(\tilde{x}). \tag{25}$$

Noise-Conditional Score Network (NCSN)

Training a score-based model at only one noise level introduces key limitations:

- At low levels, Langevin dynamics struggles to traverse modes.
- At high noise levels, the model captures only coarse structures, resulting in blurry samples.

To address these challenges, NCSN injects Gaussian noise at multiple levels into the data distribution and jointly training a noise-conditional score network (NCSN) to estimate score functions across a range of noise scales.

The goal is to train a noise-conditional score network $s_\phi(x, \sigma)$ to estimate the score function $\nabla_x \log p_\sigma(x)$. This is achieved by minimizing the DSM objective across all noise levels:

$$\mathcal{L}_{NCSN}(\phi) := \sum_{i=1}^{L} \lambda(\sigma_i) \mathcal{L}_{DSM}(\phi; \sigma_i) \tag{26}$$

and $\lambda(\sigma_i) > 0$ is a weighting function for each scale.

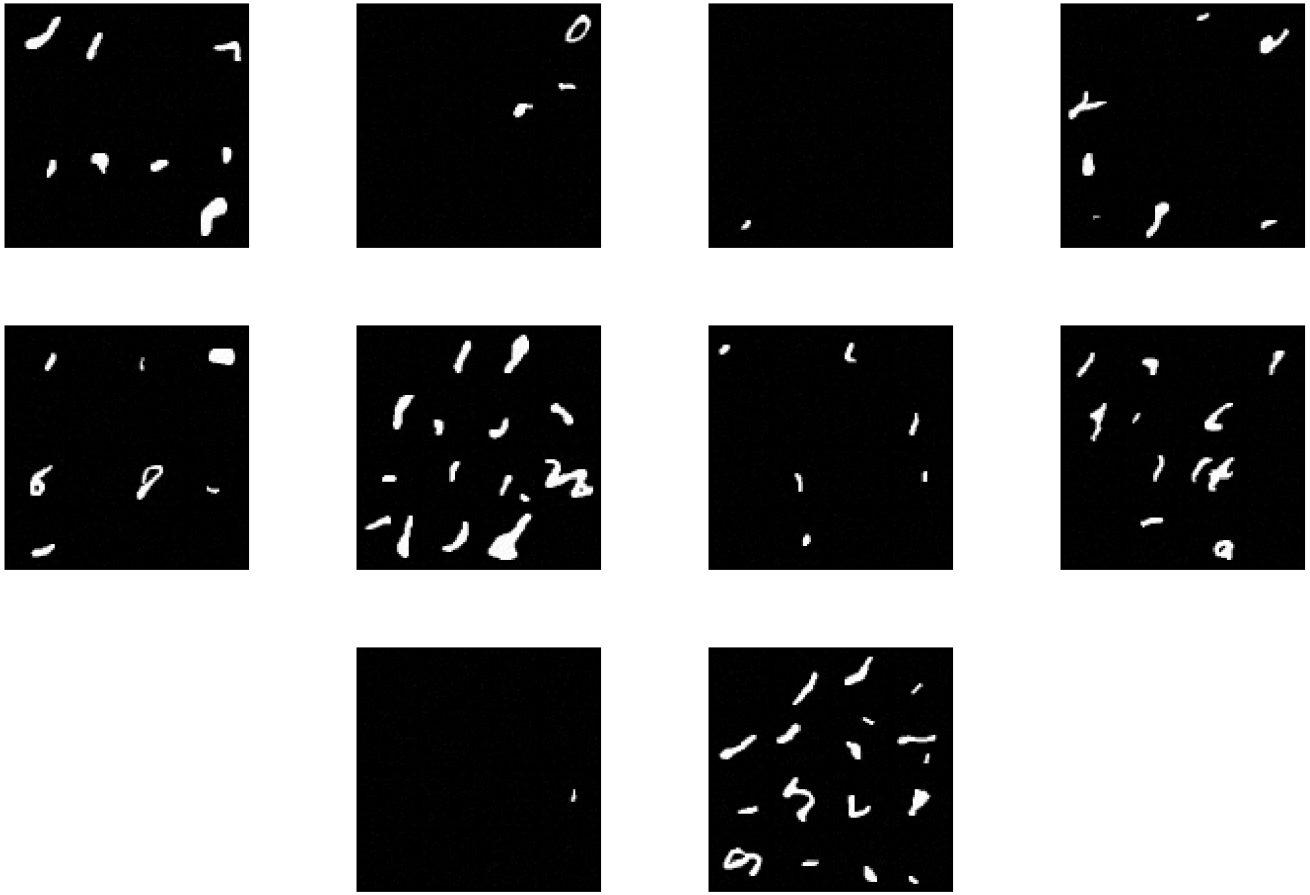
Algorithm 1 Annealed Langevin Dynamics

Input: Trained score $\mathbf{s}_{\phi^\times}(\cdot, \sigma_\ell)$, step sizes η_ℓ, and Langevin iteration budgets N_ℓ for each noise level $\ell = L, \ldots, 2$

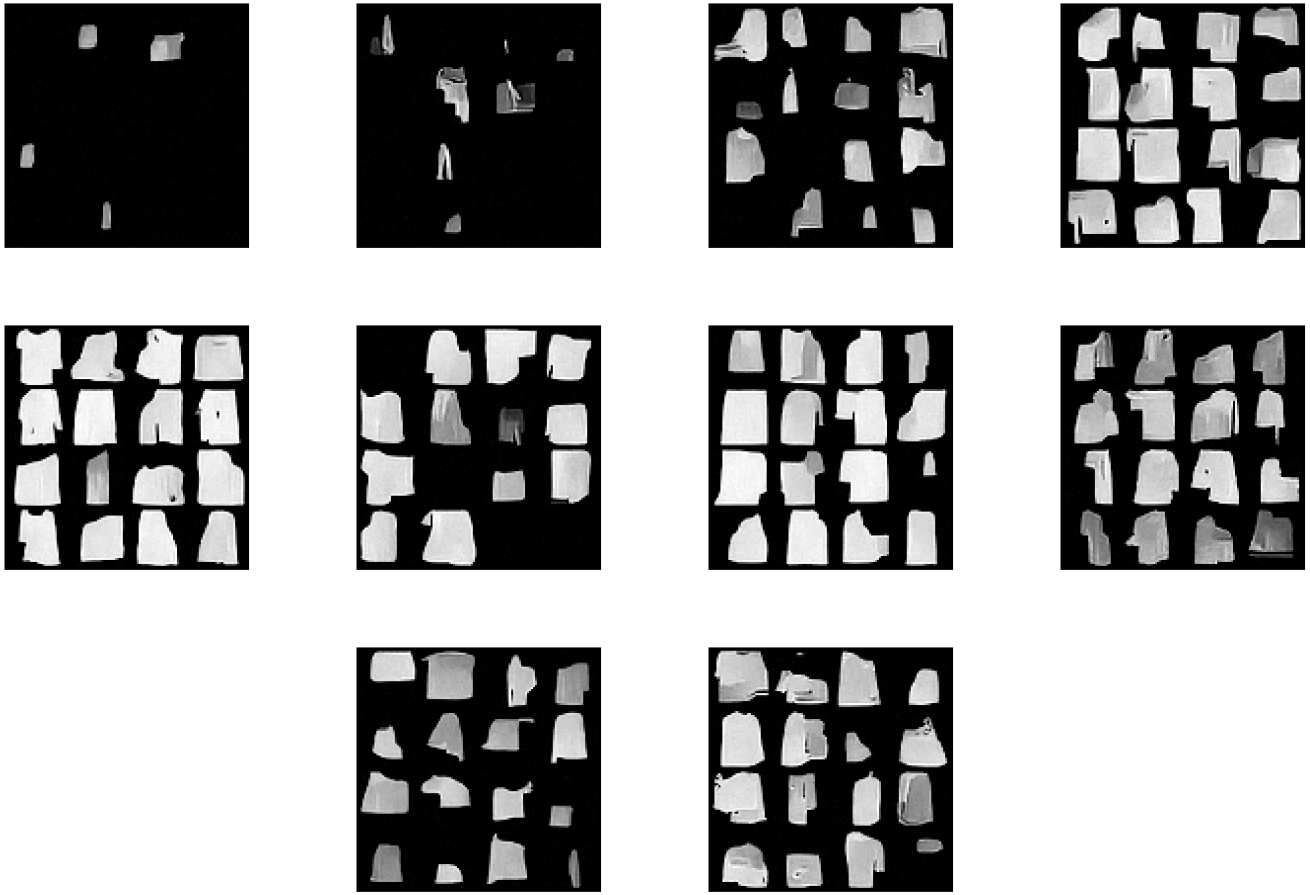
$\mathbf{x}^{\sigma_L} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
for $\ell = L, \ldots, 2$ **do**
  $\tilde{\mathbf{x}}_0 \leftarrow \mathbf{x}^{\sigma_\ell}$
  ▷ Initialize Langevin from previous noise level's output
  for $n = 0$ **to** $N_\ell - 1$ **do**
    $\boldsymbol{\epsilon}_n \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
    $\tilde{\mathbf{x}}_{n+1} \leftarrow \tilde{\mathbf{x}}_n + \eta_\ell \mathbf{s}_{\phi^\times}(\tilde{\mathbf{x}}_n, \sigma_\ell) + \sqrt{2\eta_\ell}\boldsymbol{\epsilon}_n$
  end for
  $\mathbf{x}^{\sigma_{\ell-1}} \leftarrow \tilde{\mathbf{x}}_{N_\ell}$
  ▷ Output used as initialization for next noise level
end for

Output: $\mathbf{x}^{\sigma_1}$

MNIST



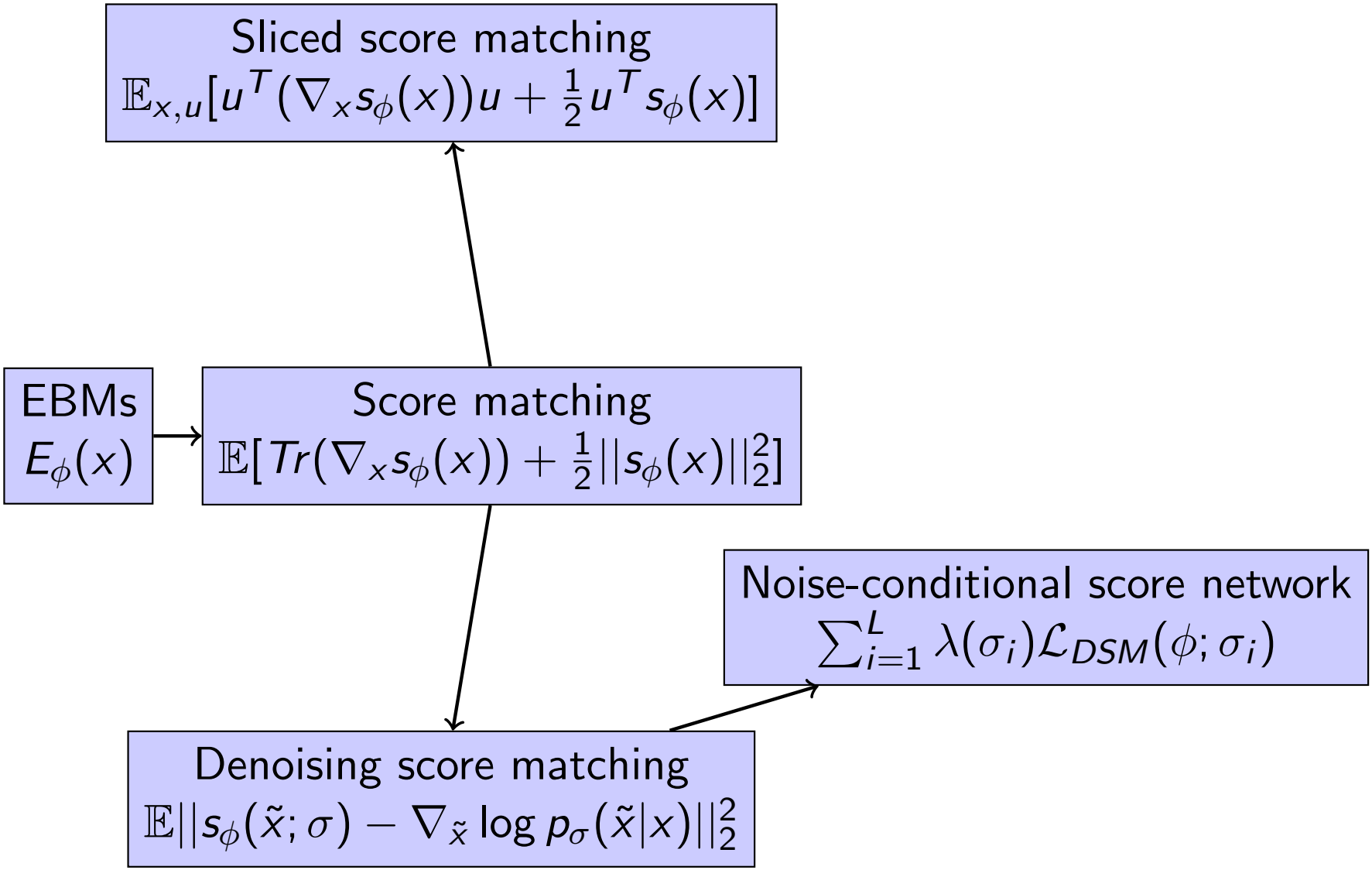
Fashion MNIST



Relationship with DDPM Loss

	NCSN	DDPM
$\mathbf{x}_{i+1}\|\mathbf{x}_i$	Derive as $\mathbf{x}_{i+1} = \mathbf{x}_i + \sqrt{\sigma_{i+1}^2 - \sigma_i^2}\boldsymbol{\epsilon}$	Define as $\mathbf{x}_{i+1} = \sqrt{1-\beta_i}\mathbf{x}_i + \sqrt{\beta_i}\boldsymbol{\epsilon}$
$\mathbf{x}_i\|\mathbf{x}$	Define as $\mathbf{x}_i = \mathbf{x} + \sigma_i^2\boldsymbol{\epsilon}$	Derive as $\mathbf{x}_i = \bar{\alpha}_i\mathbf{x} + \sqrt{1-\bar{\alpha}_i^2}\boldsymbol{\epsilon}$
p_{prior}	$\mathcal{N}(\mathbf{0}, \sigma_L^2\mathbf{I})$	$\mathcal{N}(\mathbf{0}, \mathbf{I})$
Loss	$\mathbb{E}_i\mathbb{E}_{p_{\text{data}}(\mathbf{x})}\mathbb{E}_{\boldsymbol{\epsilon}\sim\mathcal{N}(\mathbf{0},\mathbf{I})}\left[\left\|\mathbf{s}_{\boldsymbol{\phi}}(\mathbf{x}_i,\sigma_i) + \frac{\boldsymbol{\epsilon}}{\sigma_i}\right\|_2^2\right]$	$\mathbb{E}_i\mathbb{E}_{p_{\text{data}}(\mathbf{x})}\mathbb{E}_{\boldsymbol{\epsilon}\sim\mathcal{N}(\mathbf{0},\mathbf{I})}\left[\|\boldsymbol{\epsilon}_{\boldsymbol{\phi}}(\mathbf{x}_i,i) - \boldsymbol{\epsilon}\|_2^2\right]$
Sampling	Apply Langevin per layer; use output to initialize the next	Traversing the Markovian chain with $p_{\boldsymbol{\phi}^\times}(\mathbf{x}_{i-1}\|\mathbf{x}_i)$

Additionally, NCSN optimum is the true score $\nabla_{x_\sigma} \log p_\sigma(x_\sigma)$, while the Bayes optimal noise predictor under the DDPM loss is $\mathbb{E}[\epsilon|x_\sigma]$. By Tweedie's formula,

$$\nabla_{x_\sigma} \log p_\sigma(x_\sigma) = -\frac{1}{\sigma}\mathbb{E}[\epsilon|x_\sigma] \tag{27}$$

Relationship diagram